

AADISH D SOMAYAJI

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EDUCATION

Atria Institute of Technology

Bachelor of Engineering in Computer Science - Current CGPA: 9.02

2022 – Present

Sri Sri Ravishankar Vidya Mandir

Pre-University in Computer Science - 87.05%

2020 - 2021

SSLC - 84.9%

2020

SKILLS

Programming Languages : C, Python, JAVA, HTML, CSS, JavaScript, React, React Native, Tailwindcss

Soft Skills : Communication, Teamwork, Problem Solving, Decision making, Leadership

PROJECTS

- Developed a platform for AI-powered music creation that uses RNN for machine learning, allowing users to mix and compose original music from a variety of genres, including pop, jazz, rock, and classical. Future improvements include collaborative music production, for which a research paper was also written, and mood-based generation.
- Designed sensor-based specs for the visually impaired using Arduino Uno and an ultrasonic sensor, enabling proximity-based beeping for obstacle awareness. The system detects objects at specific distances and activates a connected sound device to provide audible alerts.
- Developed a smart dustbin using Arduino Uno and servo motors, featuring automated lid control that opens when waste is detected within a certain distance and closes when no obstacle is present, ensuring hands-free and hygienic disposal.
- Developed a game system using Python with Tkinter for GUI, featuring user authentication and classic games like Minesweeper, PvP Tic-Tac-Toe, and AI-based Tic-Tac-Toe, utilizing SQL for a secure database.
- Developed a personal portfolio site with React and Netlify, highlighting projects, contact info, and deployment skills in a clean, responsive layout. www.aadishds.me
- Developed a web utility called **JsonGlance** using React and Netlify that allows users to paste and instantly visualize JSON data structures in a clean, interactive, and responsive interface <https://jsonglance.netlify.app/>
- Built and deployed a chatbot web app using React and Botpress, showcasing frontend development, state management, and third-party API integration. <https://chatbot-aadish.netlify.app/>
- Developed "**AquaScope**" — an interactive 3D fish encyclopedia built with React and Vite, enabling users to explore fish species worldwide through intelligent search, detailed biological data, and immersive 3D model visualization.
- Developed "**GDHelper**" — a full-stack speech evaluation system integrating Whisper transcription, embedding-based semantic scoring, and LLM-driven feedback generation; containerized the multi-service architecture using Docker for reproducible deployment and system portability.
- Built and deployed a full-stack blog platform using React, Node.js, PostgreSQL, and Docker, featuring a public reader experience and a password-protected admin dashboard with JWT authentication, bcrypt password hashing, and rate-limited login protection; containerized with Docker Compose and hosted on AWS EC2 with HTTPS via Let's Encrypt. <https://blog.aadishds.me>
- Developed a Solar Power Forecasting system — an end-to-end ML pipeline training an LSTM neural network on ERA5 weather reanalysis data to predict hourly AC generation up to 7 days ahead; automated the inference pipeline using GitHub Actions for hourly forecast updates and deployed an interactive Streamlit dashboard on Streamlit Cloud.
- Developed "**ShopFinder**" — a location-based React Native app (using Expo) that helps users find nearby shops and manage favourites, with Firebase App Distribution for beta testing.
- Developed "**WeatherApp**" — a real-time weather tracking app built with React Native using the OpenWeatherMap API, allowing users to search cities, view current conditions, and explore detailed weather data with a clean multi-screen UI.
- Developed "**MyPokedex**" — a Pokémon information viewer built in React Native that lets users search Pokémon by name, displays shiny sprites, and shows type, weight, and height details using data from the PokéAPI.
- Developed "**FishKowner**" — a marine species exploration app built with React Native that enables users to search fish by name, view scientific and habitat details, and interact with embedded 3D models using Appwrite for data management and WebView for model rendering.
- Developed "**MyNotes**" — a full-stack note-taking app using React Native for the frontend and Express.js with MongoDB for the backend, enabling users to create, edit, and delete notes with real-time updates and clean, responsive UI.
- Developed "**Key Flow**" — a React Native typing speed test app with real-time results tracking, featuring configurable timers, accuracy and WPM calculation, and storage/display of previous tests, with a responsive and intuitive UI for both Android and iOS.

PRESENTATIONS

- Presented 'Melody Mixer: Novel Music Generation by Mixing Styles with Variational Autoencoders' at the International Conference on Computing for Sustainability and Intelligent Future (IEEE COMPSIF 2025) on March 21-22, 2025.
- Presented technical seminars on Sonar technology, BioLeaf analysis, and Photovoltaic (PV) cell innovations, showcasing applications and advancements in each field.
- Delivered a data-driven presentation on the optimal social media platforms for engagement, analyzing likes, comments, and shares to identify the most effective channels for posting.
- Presented on applications of data structures for efficient data sorting, highlighting algorithms to optimize performance and storage.

CERTIFICATION

- Earned NPTEL certification in "Introduction to Programming in C", covering fundamental programming concepts, syntax, and problem-solving techniques in C language.
- Earned NPTEL certification in "Programming, Data Structures, and Algorithms Using Python", focusing on core programming principles, efficient data handling, and algorithmic problem-solving.
- Completed NPTEL certification in "Design and Analysis of Algorithms", covering algorithm design techniques, complexity analysis, and optimization strategies.
- Received a certificate of participating in "Celebrating Tech @Bangalore DC" held at Infosys Bangalore on October 23rd, 2024 where we presented our project idea named "The Wild Sense".
- Received a certificate of participation in "NVIDIA GeForce Day" event held at Bangalore on July 27th, 2023 for participating in multiple workshops held during the event.
- Received a course completion certificate for the course "C Programming 101" on November 24th, 2023 organized by Infosys.
- Received third place at the event "Rapid Quiz" held at Atria Institute of Technology on the occasion of National Science day between February 23rd and 27th, 2023.

PUBLICATIONS

- *Melody Mixer: Novel Music Generation by Mixing Styles with Variational Autoencoders* (Paper ID: 354). Published in the **Proceedings of the 2025 IEEE International Conference on Computing for Sustainability and Intelligent Future (COMP-SIF)**.
- Published a book chapter titled "AI-Driven Marine Species Identification and 3D Model Generation: A Cloud-Native Approach for Sustainable Aquatic Science" in **Sustainable Approaches in Aquatic Science (ISBN: 978-81-994425-4-2), October 2025**.

POSITION AND RESPONSIBILITIES

- TEDxAtriaIT – Hospitality Lead, responsible for connecting with speakers and organizing all aspects of the event.
- Organiser for Treasure Hunt - Organized a college-wide Treasure Hunt with a team of four, incorporating technical and tricky quizzes as challenges throughout the event.